

Unveiling U.S. Bird Biodiversity

A Geospatial and Machine Learning Analysis of eBird Observation Data

How does bird observation data vary across an urbanizing country — and what does that tell us about which species adapt to human-altered environments?

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Data source: eBird API, Cornell Lab of Ornithology · 10 U.S. states

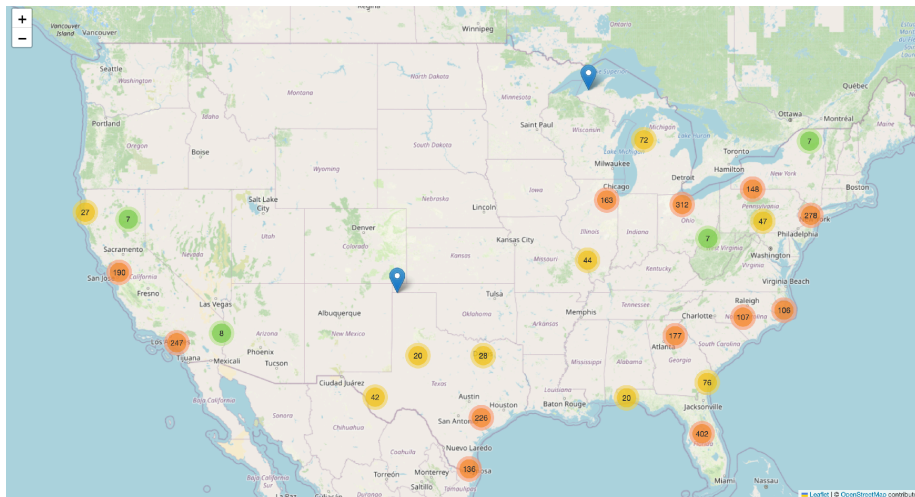
Overview

Bird populations are widely used as indicators of overall ecosystem health, making them a useful lens for studying how urbanization affects biodiversity. This project pulls real-time observation data from the eBird API — one of the largest citizen-science biodiversity projects globally — across ten U.S. states (NY, CA, TX, FL, PA, IL, OH, GA, NC, MI), then explores geographic, seasonal, and species-level patterns in how bird activity varies with region and human development.

This analysis is framed as a foundation, not a final answer: a first pass at the kinds of geographic and temporal patterns worth pursuing with more rigorous urbanization-metric data in future work.

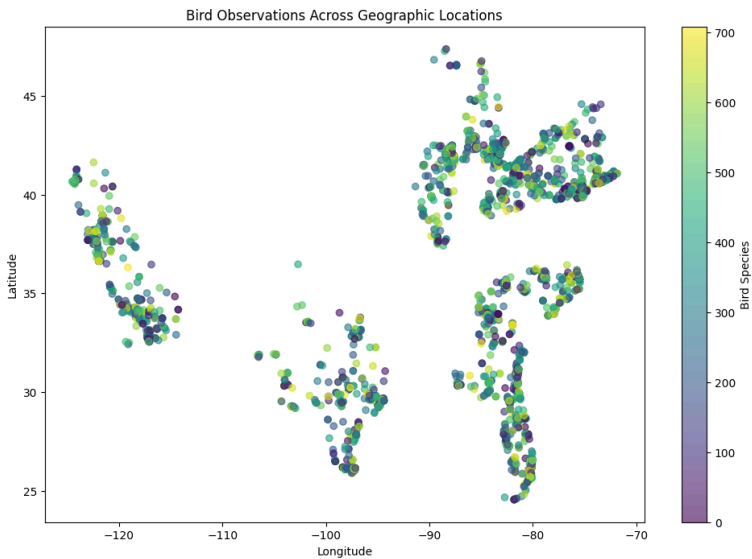
Geographic Distribution

An interactive Folium map (screenshot below) clusters observations across the ten sampled states, giving a first read on where activity concentrates.



Interactive map of clustered bird observations across the ten sampled states.

A latitude/longitude scatter plot, colored by species, gives a complementary view of the same spatial spread at the individual-observation level:



Observation locations colored by species.

Species Prevalence

The ten most frequently observed species across the full sample:

Top 10 Most Frequently Observed Bird Species

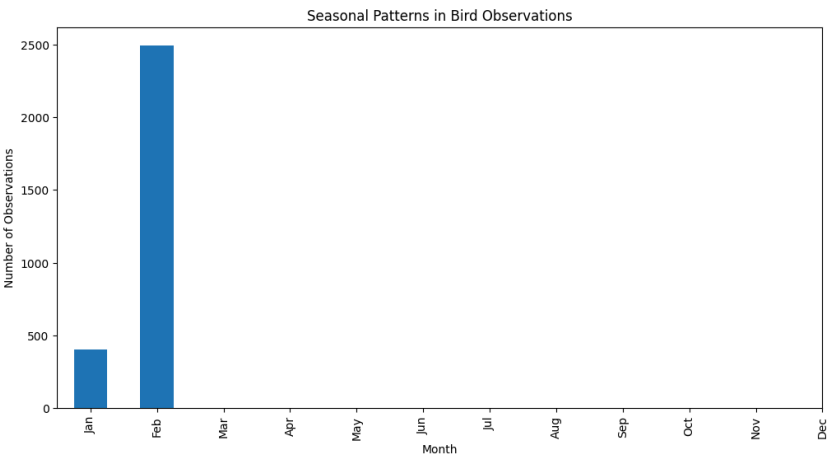
Species	Observations
Northern Mockingbird	10
Pileated Woodpecker	10
White-Winged Scoter	10
Long-Tailed Duck	10
Ring-Necked Duck	10
Killdeer	10
Red-Winged Blackbird	10
Common Goldeneye	10
Peregrine Falcon	10
Lesser Black-Backed Gull	10

Top 10 most frequently observed species, full sample.

Species that show up most often across varied locations are plausible candidates for birds that are particularly adaptable to human-altered environments — a signal worth following up with dedicated habitat data in future work, rather than a conclusion this dataset alone can support.

Temporal Patterns

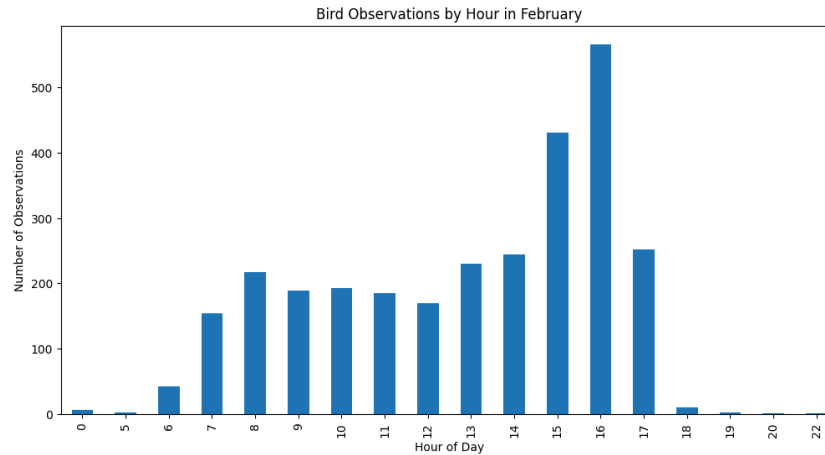
Monthly Observation Counts



Observations by month.

Because the underlying pull covers a recent 30-day observation window, this monthly view mostly reflects when the data was collected rather than a true multi-year seasonal cycle — a genuine limitation, and one of the clearest opportunities for future work: a longitudinal pull across a full year (or several) would be needed to say anything firm about seasonal bird activity.

Hourly Observation Patterns



Observations by hour of day.

The hourly distribution reflects a mix of actual bird activity patterns and observer behavior (i.e., when birdwatchers are out logging sightings) — both signals are entangled in citizen-science data like this, which is itself a useful methodological takeaway for anyone working with eBird-style datasets.

Habitat Classification

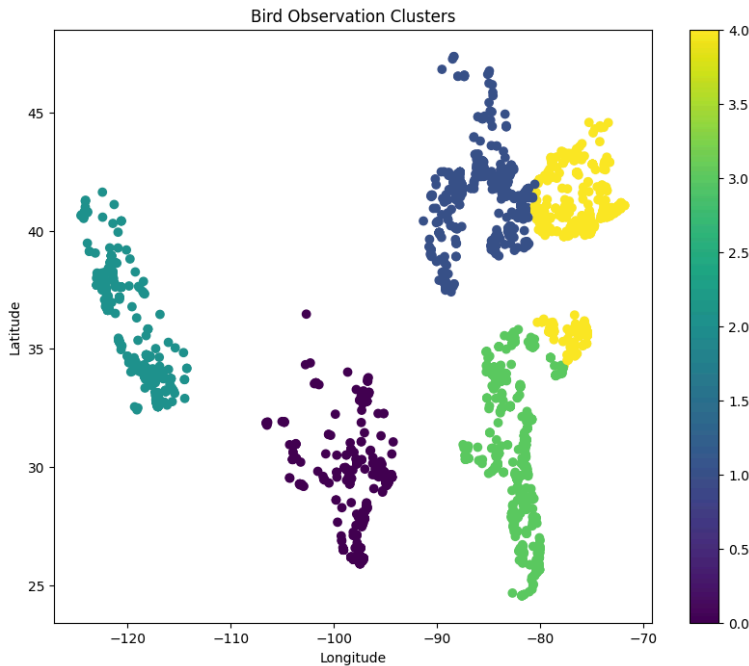
Observations were grouped into four broad regional categories based on latitude and longitude (Northern, Southern, Western, Eastern), then examined for the top species within each region:

Region	Top Species (tied at top observation count)
Northern	Bufflehead, Common Goldeneye, Great Egret, Northern Shrike, Peregrine Falcon
Eastern	American Coot, Common Loon, Hooded Merganser, Northern Shoveler, Purple Finch
Southern	American Avocet, American Bittern, American Crow, American Goldfinch, American Oystercatcher
Western	Allen's Hummingbird, American Barn Owl, American Robin, Black-Chinned Sparrow, Bonaparte's Gull

Regional species lists differ meaningfully — Northern and Eastern regions lean toward waterfowl and wetland species, while Western and Southern lists show more variety. Worth noting honestly: within this particular data pull, top species within each region were frequently tied at low observation counts, reflecting the limited sample size of a 30-day window rather than a definitive species ranking. This is a pattern-finding exercise, not a statistically powered comparison — a good next step would be pulling a full year of data per region to get stable rankings.

Geographic Clustering

K-means clustering was applied to observation coordinates to identify geographic hotspots of bird activity, independent of the manual region labels used above:



K-means clusters of observation locations (K=5).

Each cluster's dominant species differ, which is a useful cross-check against the manual habitat classification — clustering lets the data define its own regional groupings rather than relying on a fixed lat/lng rule, and the two approaches can be compared for agreement in future iterations of this project.

Conclusion & Future Directions

This project demonstrates a data-driven approach to exploring the relationship between urbanization and bird biodiversity across a wide geographic area, using freely available citizen-science data. The methods here — API-based data collection, interactive mapping, temporal pattern analysis, and geographic clustering — form a reusable pipeline that could be extended to other regions or species groups.

The most valuable next steps identified during this analysis:

- Longitudinal data collection across a full year (or multiple years) to capture genuine seasonal trends rather than a single 30-day snapshot.
- Fine-grained urbanization metrics (building density, green space coverage) layered onto observation locations, replacing the current simple lat/lng habitat classification.
- Species-specific deep dives on the handful of species that show up as consistently adaptable across regions.
- Climate change indicators layered in, since shifting climate and urbanization likely interact in how they affect bird populations.

As cities continue to expand, understanding how different species respond to urban environments will matter for both conservation planning and urban design. This project is a first step toward that understanding, built to be extended rather than treated as a finished answer.

Data Source

- eBird. (2023). eBird: An online database of bird distribution and abundance [web application]. eBird, Cornell Lab of Ornithology, Ithaca, New York. Available: ebird.org.